

A Splined Outer Mould Line for the IV-1

The same two-stage interceptor, with its nose and interstage revolved from B-splines inside nTop

Prepared with nTop Automate 5.53.2 / 5.54.0 and SUAVE 2.5.2. All geometry is authored programmatically. Every number is read from the recorded run artefacts under runs/IV-1_spline and runs/IV-1_ogive_baseline.

Summary

This report describes one change to the IV-1 interceptor. The stage-2 nose and the interstage shoulder are now true revolved B-splines. They were a tangent ogive and a conical shoulder. Nothing else changed. The requirements, the physics modules, the mass model and the commanded pitchover angle are the same.

Both vehicles meet all 15 constraints. Both reach the same 100 mile slant range, because that is the constraint the trajectory is flown to. The spline reaches it 4.44 km higher: 19.95 km becomes 24.39 km, a rise of 22.3 percent. Intercept Mach rises from 4.004 to 4.057. Launch mass falls from 592.24 kg to 591.05 kg.

The altitude gain has a price, and this report does not bury it. Aerodynamic lateral acceleration at intercept falls by 48.5 percent, from 7.45 g to 3.84 g. The reason is simple. A higher intercept sits in thinner air, and aerodynamic manoeuvre needs dynamic pressure. Requirement A11 is therefore met almost entirely by the attitude-control thrusters. Section 5 states the trade in full.

Quantity	Unit	Tangent ogive	Spline	Change	Change [%]
Launch mass	kg	592.24	591.05	-1.19	-0.20
Slant range at intercept	km	160.93	160.93	-0.00	-0.00
Intercept altitude	km	19.95	24.39	+4.44	+22.28
Intercept Mach	-	4.004	4.057	+0.053	+1.32
Time to intercept	s	157.8	153.7	-4.1	-2.60
Dynamic pressure at intercept	kPa	62.57	32.24	-30.33	-48.47
Mass at intercept	kg	250.97	249.78	-1.19	-0.47
Lateral g, aerodynamic	g	7.451	3.837	-3.613	-48.50
Lateral g, attitude control	g	15.865	15.865	+0.000	+0.00
Lateral g, A11 figure	g	15.865	15.865	+0.000	+0.00
CN_max at intercept	-	3.228	3.211	-0.017	-0.52
Peak dynamic pressure	kPa	304.58	307.94	+3.36	+1.10
Apogee	km	37.94	39.21	+1.27	+3.35
Attitude-control thrust	kN	39.04	38.86	-0.18	-0.47
Mass jettisoned at separation	kg	31.26	31.26	+0.00	+0.00

Table 1. The two converged interceptors, side by side. Both fly the same 38 degree commanded pitchover. The A11 figure is the GREATER of the aerodynamic and the attitude-control capability, never their sum, because commanding an aerodynamic turn and a divert at the same time is a control problem this model does not represent.

1. What changed, and what did not

Design variable	Tangent-ogive baseline	Spline study
nose_shape	tangent_ogive	spline
nose_blend	0.00	1.00
interstage_shape	cone	spline
interstage_blend	0.00	1.00
Commanded pitchover [deg]	38.0	38.0

Table 2. The whole difference between the two studies. nose_blend is 1.0 here, which is the full slender-body drag optimum. That differs from the SV-1 spline study, where the sizing search stopped short of the optimum at 0.7. The IV-1 shape was set rather than searched, so this study makes no claim about an interior optimum on this vehicle.

The stage geometry, the propellant loads, the strakes, the fins and the attitude-control pack are unchanged. The attitude-control thrust is re-solved to its own fixed point at every design point, so it moves a little: 39.04 kN becomes 38.86 kN. That is a consequence of the lighter vehicle, not an input.

The nose is a cubic B-spline with 9 control points on a nose of length over radius 6.00. Its wave-drag shape ratio against the tangent ogive is 0.87454. The ogive baseline reports 1.00000 by construction, exactly 1.0, which is the check that the shape model changes nothing at the baseline shape.

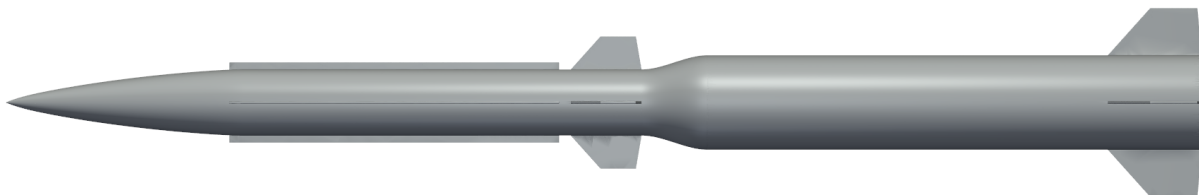


Figure 1. The converged IV-1, rendered from the STL that nTop exported at this design point. Nose tip at the left. The splined payload-stage nose runs back to the strakes, the four stage-2 tail fins sit at the interstage joint, and the SPLINED INTERSTAGE FLARE is the smooth expansion aft of them, where the tangent-ogive configuration carries a straight conical shoulder. The booster and its four tail fins complete the stack. Overall length 5.08 m.

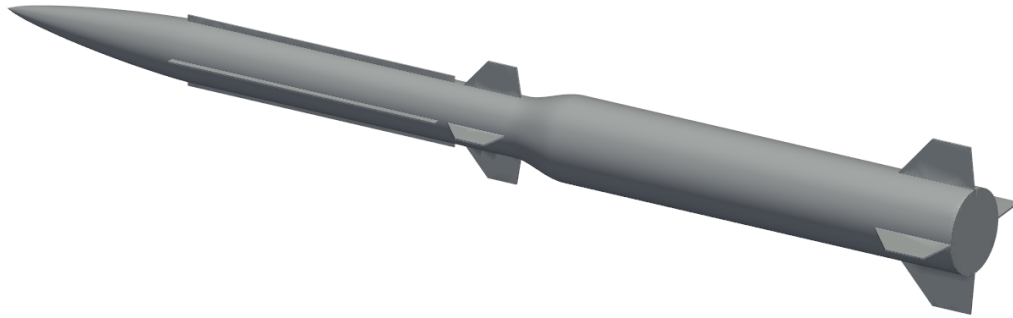


Figure 2. The same vehicle from an oblique view, which shows the four strakes and the two cruciform fin sets as separate bodies rather than as an outline.

nTop revolves the spline itself: no chord polygon, no discretisation error. The shape ratio multiplies the calibrated wave-drag term only.
Source: runs/IV-1_spline/figures/evidence_iv1_spline.json, section wavedrag.

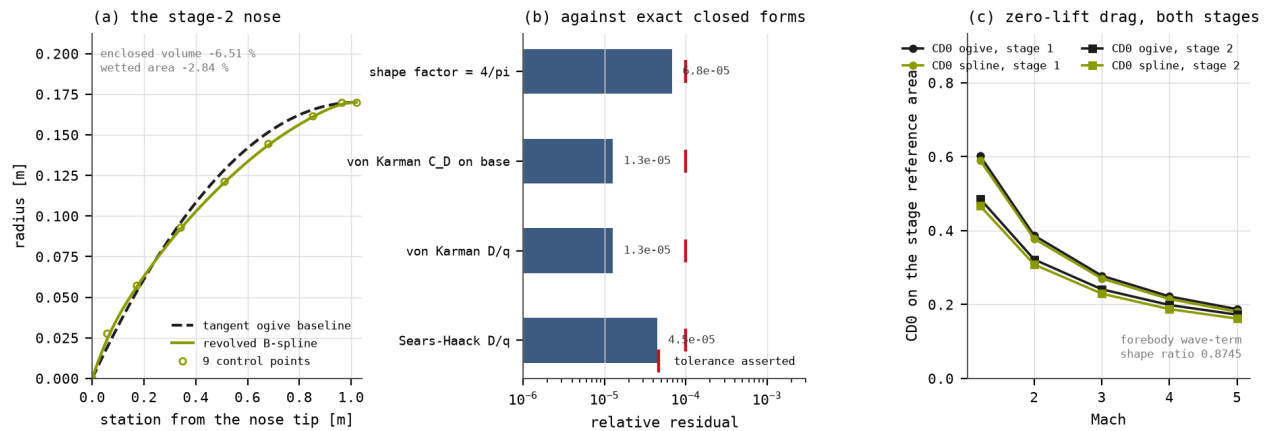


Figure 3. The revolved spline. Panel (a) is the stage-2 nose and the nine control points the notebook computes. Panel (b) is the closed-form validation of the wave-drag model. Panel (c) is the zero-lift drag it moves, in both flight configurations.

2. Why the shape can change anything at all

The forebody wave-drag term in this repository is the Bonney correlation. It is a function of length over diameter alone. It cannot tell one nose from another at fixed fineness. A spline study run against that model would have reported no change, for the wrong reason.

rocketgen/sizing/wavedrag.py supplies the missing sensitivity. It computes the linearised slender-body wave drag of an area distribution through the Glauert series. It then reports a dimensionless RATIO against the tangent ogive of the same fineness. The ratio multiplies the correlation value. So the calibrated level and the Mach dependence stay with the correlation. Only the shape effect comes from linear theory. At the ogive control values the ratio is exactly 1.0.

Check	Closed form	Model	Relative residual
Sears-Haack body drag	$128 V^2 / (\pi L^4)$	0.968990	4.5e-5
von Karman ogive drag	series against the closed form	0.00267257	1.3e-5
von Karman C_D on base area	$(d / L)^2$	0.02777813	1.3e-5
Optimum shape factor	$4 / \pi$	1.27332596	6.8e-5

Table 3. The wave-drag model against exact closed forms from outside this repository. Every test asserts a tolerance of 1.0e-4. The residuals above are measured for this report rather than transcribed from the test file, so they are what the code achieves and not what it is allowed.

Stations in the check table	Relative residual
501	1.7e-2
1001	2.6e-3
2001	2.0e-4
4001	4.5e-5
8001	1.7e-5

Table 4. The Sears-Haack residual belongs to the CHECK TABLE, not to the model. The Sears-Haack profile has an infinite slope at both ends, so the central-difference derivative converges slowly on it. Refining the table drives the residual down. A single number could not have said this.

Two further checks support the series. The Glauert series constant of pi over four was compared against direct double integration of the same integral. The deleted diagonal converges slowly, so what is measured is convergence onto the series: the residual falls from 4.8e-2 at 250 points to 1.5e-2 at 1000. Separately, the von Karman ogive was confirmed as the constrained optimum rather than assumed to be. Adding any higher Glauert mode at fixed base area raises the drag in all 4 perturbations tried. A 9-point spline recovers 86.2 percent of the distance from the tangent ogive to that bound.

Mach	Stage	CD0 ogive	CD0 spline	Change [%]	Wave share of CD0
1.2	1	0.60224	0.59003	-2.03	14.8 %
1.2	2	0.48586	0.46703	-3.88	28.1 %
2.0	1	0.38650	0.37772	-2.27	16.5 %

Mach	Stage	CD0 ogive	CD0 spline	Change [%]	Wave share of CD0
2.0	2	0.32167	0.30811	-4.22	30.3 %
3.0	1	0.27772	0.27007	-2.75	20.2 %
3.0	2	0.24093	0.22914	-4.89	35.5 %
4.0	1	0.22167	0.21446	-3.25	24.0 %
4.0	2	0.19851	0.18742	-5.59	40.9 %
5.0	1	0.18725	0.18026	-3.74	27.7 %
5.0	2	0.17241	0.16166	-6.23	46.0 %

Table 5. What the shape does on this vehicle, at 12 km and 2 degrees of angle of attack. Stage 1 is the full stack on the booster reference area. Stage 2 is the surviving stage on its own area. A coefficient computed on the wrong area is silently wrong by the diameter ratio squared, so the two configurations are reported separately and never mixed.

3. How the spline is built in nTop

nTop revolves the spline itself. There is no chord polygon and no sampling error. The notebook computes the control points from live inputs, builds a profile from a list of curves, and revolves that profile. The chain is four blocks:

```
spline_by_control_points<list<point>,integer>[5.20.0]
core.list<curve_interface>
profile_from_curves<list<curve_interface>,vector>[5.20.0]
revolve<new_profile,axis,real>[5.20.0]
```

The profile is a curve LIST. It mixes spline segments with two_point_line segments, so the straight runs stay exactly straight and the corners stay sharp. The axial fractions of the control points are the Greville abscissae, which makes the axial station an exact linear function of the spline parameter.

None of those blocks is in the vendored block universe. It carries 853 signatures and 0 of the 4 blocks in the chain. The universe is incomplete, and its silence is not evidence that nTop cannot do something. An earlier attempt sampled the spline into a chord polygon, because the universe listed no route from a curve to a revolvable profile and that was taken as proof no route existed. It was not proof. All four blocks go through the raw-block escape hatch. docs/NTOP_NOTES.md section 25 records the four encoding traps.

Body	nTop volume [m^3]	Exact integral [m^3]	Relative error	Measurement wall time [s]
Stage 2, revolved cubic B-spline, exact integral	0.171870	0.171835	+0.0205 %	71.8
Stage 2, tangent ogive, quadrature	0.175058	0.175074	-0.0091 %	104.6

Table 6. The measured stage-2 solid against the exact integral of the shape that built it. For the spline that closed form is exact, because nTop revolves the same B-spline. There is no discretisation error available to hide behind. The splined nose gives up 6.51 percent of enclosed volume and 2.84 percent of wetted area against the ogive.

Notebook	Recipe body nodes
IV-1 stack, splined nose and interstage	455
IV-1 stack, tangent ogive and cone	540

Table 7. Notebook size, counted as nodes in the recipe body. The spline notebook is the smaller of the two, because a revolved spline replaces the sampled polygon and the arithmetic that built it.

One notebook that reports several bodies must namespace its output keys. This one emits per-stage prefixes, so the generic output parser is not safe to call on it and a dedicated reader returns a dictionary keyed by stage. It also states which frame every measured station is in. The structural centre of gravity is reported both in stage-local coordinates and in stack coordinates, because the mass model adds the stage offset itself. Getting that wrong moves a centre of gravity by metres and nothing crashes.

4. The converged interceptor

Constraint	Value	Sense	Limit	Margin [%]	Met
A6 max diameter [m]	0.4200	<=	0.4200	+0.00	yes
stage 1 grain closes	1.0000	>=	1.0000	+0.00	yes
stage 2 grain closes	1.0000	>=	1.0000	+0.00	yes
A2 slant range [m]	160,934.0000	>=	160,934.0000	+0.00	yes
A7 stacked length [m]	5.2800	<=	5.4000	+2.22	yes
A11 lateral g	15.8649	>=	15.0000	+5.77	yes
stage 2 vol loading	0.8922	<=	1.0000	+10.78	yes
A10 q_max [Pa]	307,939.4728	<=	350,000.0000	+12.02	yes
A4 intercept Mach	4.0571	>=	3.0000	+35.24	yes
A8 launch mass [kg]	591.0502	<=	1,400.0000	+57.78	yes
stage 1 vol loading	0.3747	<=	1.0000	+62.53	yes
A3 intercept alt [m]	24,392.1617	>=	15,000.0000	+62.61	yes
grain L/D stage 1	2.6079	<=	8.0000	+67.40	yes
A12 s1 burnout alt [m]	4,683.3434	<=	20,000.0000	+76.58	yes
grain L/D stage 2	1.7971	<=	8.0000	+77.54	yes

Table 8. All 15 constraints the loop checks, at the converged spline design, sorted by margin. Say how many were checked, not only that they were met. A requirement that is not in this list is not tested. The IV-1 study shipped with two holes in this list once: grain closure was not gated, and the stage-2 static margin was never added. Grain closure is now the last two rows.

Bars are the spline result. Orange ticks are the tangent-ogive baseline at the same pitchover angle. Both meet all fifteen.
Source: runs/IV-1_spline/figures/evidence_iv1_spline.json, section constraints.

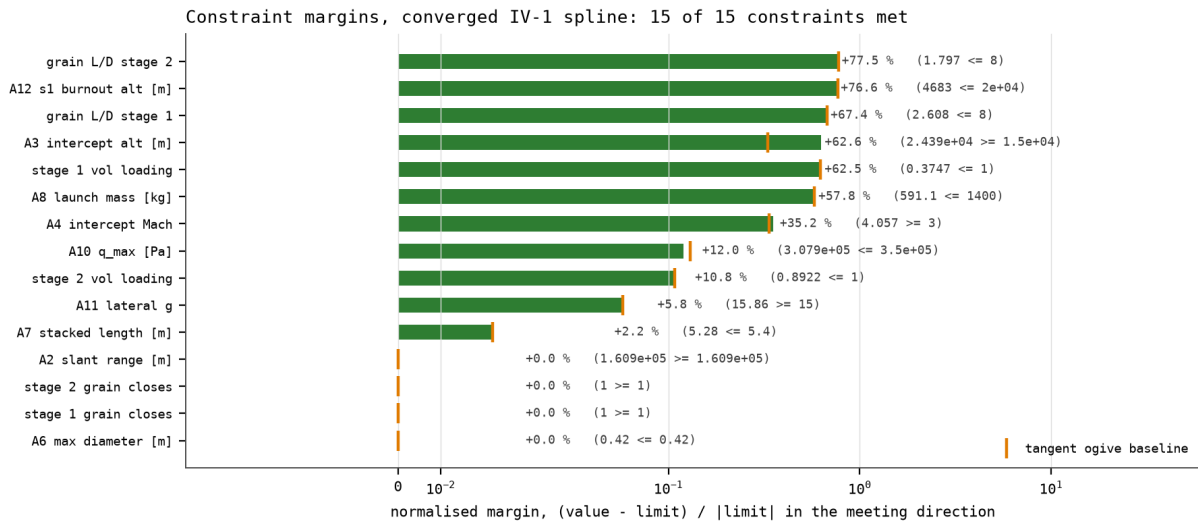


Figure 4. The same margins, drawn. Orange ticks are the tangent-ogive baseline at the same pitchover angle. The two shapes trade margin between A3 and A11.

The trajectory carries 7,688 recorded samples over 153.7 s. Apogee is 39.21 km at $t = 98.5$ s. Peak dynamic pressure is 307.9 kPa at 4.63 km and Mach 2.78. The integrator makes every event time a hard step boundary, so the mass bookkeeping through staging closes to machine precision.

The same interceptor flown twice. Only the nose and interstage shape family differs. Both fly the same 38 degree pitchover.
Source: runs/IV-1_spline/figures/evidence_iv1_spline.json.

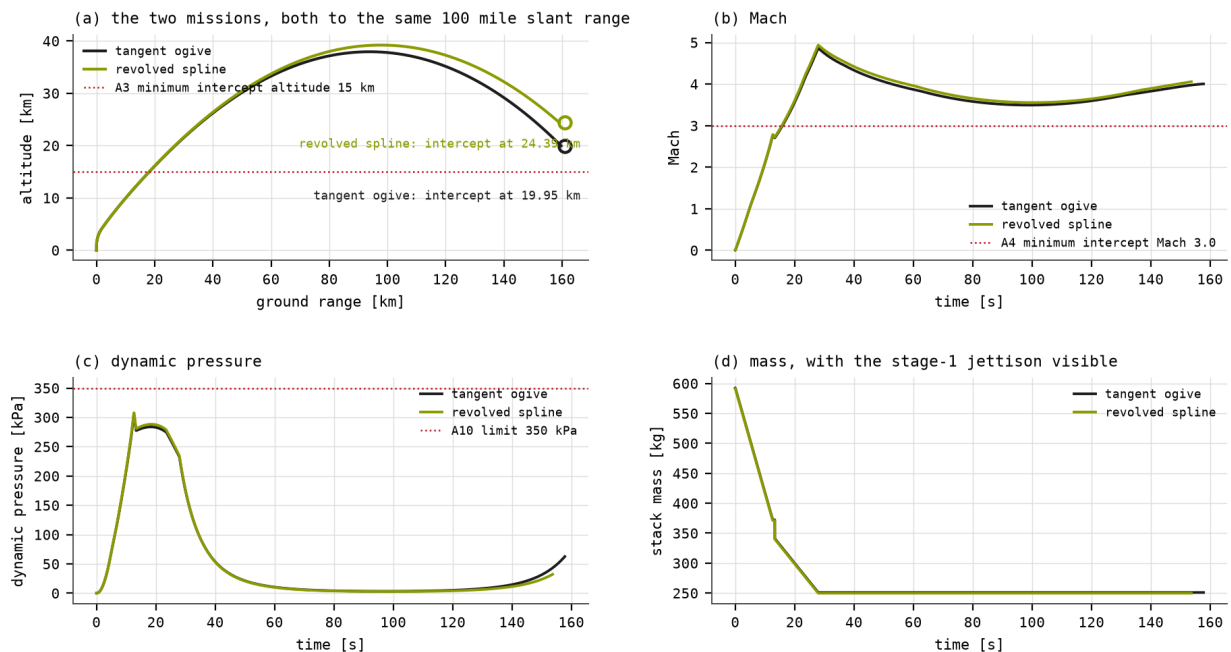


Figure 5. The two flown missions. Both reach the same 100 mile slant range, because that is what the trajectory is flown to. The spline arrives higher, faster and in thinner air.

Stage	Item	Mass [kg]	Station [m]	Provenance
1	Propellant	220.00	4.084	analytic
1	Airframe structure and surfaces	39.96	4.171	ntop_measured
1	Dry-mass contingency	11.11	4.149	correlation
1	Motor hardware not modelled	7.86	4.084	correlation
1	Motor insulation	6.76	4.084	analytic
1	Motor case	4.56	4.084	analytic
1	Nozzle assembly	1.47	5.150	correlation
1	Igniter	1.10	3.127	analytic
2	Propellant	90.00	2.070	analytic
2	Payload	75.00	1.540	requirement
2	Airframe structure and surfaces	23.81	1.530	ntop_measured
2	Radome	11.19	0.459	analytic
2	Dry-mass contingency	7.90	1.372	correlation
2	Motor insulation	3.56	2.070	analytic
2	Motor case	1.87	2.070	analytic
2	Nozzle assembly	1.76	2.280	correlation
2	Motor hardware not modelled	1.25	2.070	correlation
2	Igniter	0.45	1.806	analytic
0	Interstage	3.02	2.540	analytic
	ATTITUDE-CONTROL PACK	78.41		requirement
	TOTAL	591.05		

Table 9. The group-weight statement, with the provenance of every line. 12.44 percent of the launch mass is measured in nTop, which is 63.76 kg of airframe and surfaces. The stage totals are 292.82 kg for stage 1, 216.80 kg for stage 2 and 3.02 kg for the interstage. 31.26 kg leaves the vehicle at separation.

5. The trade: a higher intercept costs aerodynamic manoeuvre

This is the finding a reader must not miss. The spline reaches the required slant range at a higher altitude. Higher altitude means thinner air. Dynamic pressure at intercept falls from 62.57 kPa to 32.24 kPa, a drop of 48.5 percent. Aerodynamic lateral acceleration is proportional to dynamic pressure, so it falls by very nearly the same amount: 7.45 g becomes 3.84 g. The maximum normal-force coefficient barely moves, from 3.228 to 3.211, which confirms that the loss is dynamic pressure and not aerodynamics.

Contribution to A11	Tangent ogive [g]	Spline [g]	Change [%]
Aerodynamic	7.451	3.837	-48.50
Attitude control	15.865	15.865	+0.00
A11 figure, the greater of the two	15.865	15.865	+0.00

Table 10. How requirement A11 is met. For the tangent ogive the aerodynamic contribution is 47.0 percent of the A11 figure. For the spline it is 24.2 percent. A11 is now met by the thrusters almost alone. The requirement still passes. The margin behind it is thinner, because a thruster has a finite total impulse and the air does not.

That matters because the attitude-control pack is not free. It weighs 78.41 kg and carries 116.6 kN.s of total impulse for a burn of a few seconds. A design that leans harder on it has less divert authority in reserve for the endgame. This report records that as a real cost of the altitude gain, not as a rounding detail.

6. The pitchover sweep

The commanded pitchover angle is not a geometry input. Sweeping it changes the pitch programme and the attitude-control thrust, and nothing else. One measurement set per shape therefore serves every angle, so the sweep below is a genuine nTop-coupled sweep and not an analytic stand-in.

Pitchover [deg]	Shape	Slant range [km]	Intercept altitude [km]	Mach	Aerodynamic g	Feasible
32	ogive	151.66	-0.00	1.763	27.85	no
32	spline	159.26	0.00	1.882	31.80	no
34	ogive	160.93	2.90	2.370	35.29	no
34	spline	160.93	8.67	3.338	31.03	no
36	ogive	160.93	12.05	3.698	22.23	no
36	spline	160.93	16.92	3.977	11.85	yes
38	ogive	160.93	19.93	4.004	7.47	yes
38	spline	160.93	24.38	4.057	3.85	yes

Table 11. The sweep covers 4 angles. The spline is feasible at 2 of them. The ogive is feasible at 1. The spline opens a feasible angle the ogive does not have: at 36 degrees the spline meets every constraint, and the ogive fails A3 with an intercept altitude of 12.05 km against a 15 km minimum. That is the clearest single statement of what the shape bought.

The pitchover sweep, with the geometry measured. The pitchover angle is not a geometry input, so one measurement set serves every angle.
Source: runs/IV-1_spline/figures/evidence_iv1_spline.json, section pitchover.

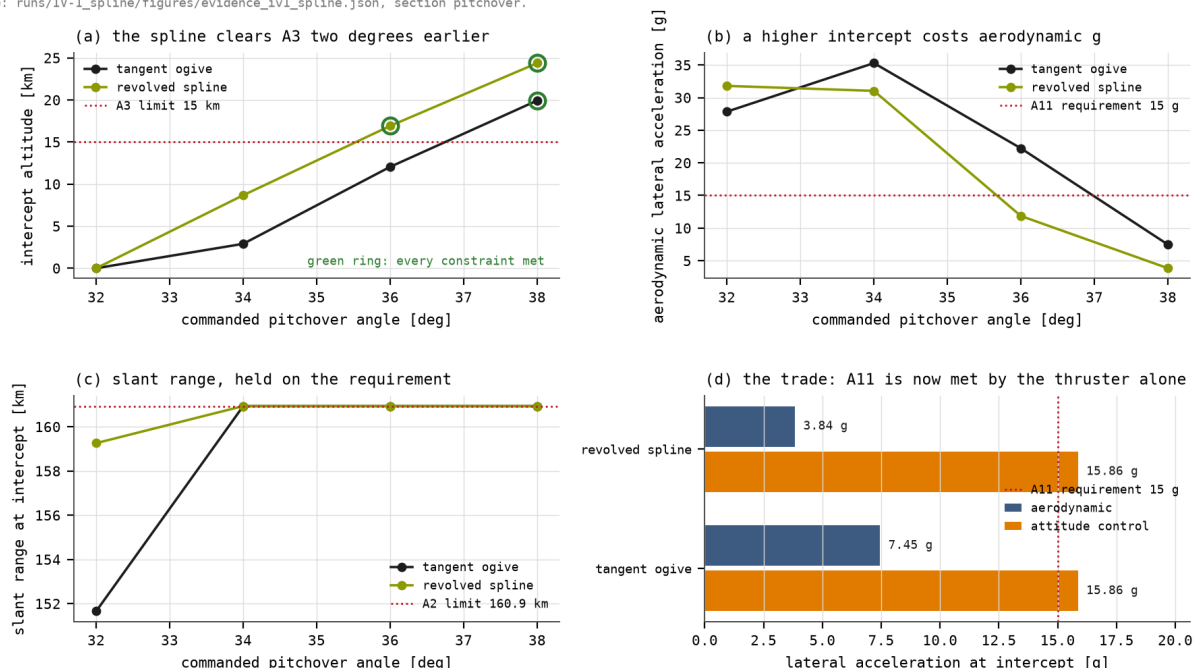


Figure 6. The pitchover sweep. Panel (a) shows the spline clearing A3 two degrees earlier. Panel (b) shows what that costs in aerodynamic g. Panel (d) shows how A11 is actually met at the converged point.

7. An open defect: the drag calibration is not applied to IV-1

The zero-lift drag calibration factor is validated against 23 Basic Finner free-flight shots. Its value is 1.171. The SV-1 loop applies it at the loop boundary, through CalibratedAero. **The IV-1 study does not.** scripts/iv1_converge.py builds the stack aerodynamics directly and never wraps it. So every IV-1 result, including both results in this report, runs about 15 percent low on zero-lift drag against that data set.

Item	Value
Calibration factor available	1.171
Applied to IV-1	NO

Table 12. The open defect, recorded in the machine-readable evidence file so it cannot be forgotten. A calibration that exists in the repository, is tested, and is silently absent from half the results is worse than not having one. A reader sees the mechanism and assumes it applied.

The effect on the COMPARISON in this report is smaller than the effect on either absolute result. Both vehicles are uncalibrated in the same way, and the shape ratio multiplies the same term in both. The comparison is therefore still a fair one. The absolute intercept altitudes, Mach numbers and ranges are optimistic by an amount this study has not quantified.

8. Limitations

The source registry holds 161 entries once every owning module is imported. 57 of them are flagged. A flagged entry is one whose source string contains the word guess, or the words modelling choice, approximation or assumption. Every empirical constant in this repository carries a source string, and a test asserts that a guess says so.

The registry is filled at import time, so it is only complete once every owning module has been imported. The evidence collector imports all sixteen owning modules before it reads the registry, for that reason. A report that read the registry too early once listed 37 sources instead of 70, and under-reported its own limitations by a factor of four.

Owning module	Registered sources
aero	23
propulsion	22
aero_iv1	14
ntopgen.stack_notebook	14
propulsion_iv1	13
trajectory_iv1	13
trajectory	11
masses	10
wavedrag	8
ntopgen.rocket_notebook	8
oml_spline	7
config_iv1	6
masses_iv1	5
config	4
atmosphere	3
TOTAL	161

Table 13. Where the registered sources live. The config module is not listed as an owner of the other modules' entries, because the registry IS the config module: attributing to it would attribute every key in the repository to it.

Every flagged source

Flagged source	What it says
aero_isentropic_compression_turn	Pressure rise through a compressive turn taken as an exact isentropic Prandtl-Meyer compression, $nu_2 = nu_1 - \Delta$, with $C_p = (2/(\gamma M_1^2))(p_2/p_1 - 1)$. Same gas dynamics and the same reference as the boattail expansion, Liepmann and Roshko, Elements of Gasdynamics (1957), run in the opposite direction. APPROXIMATION: the entropy rise across the real oblique shock is not modelled, and when Δ exceeds the available Prandtl-Meyer angle the shock detaches and the value is clamped at the sonic result, which UNDERSTATES the detached-shock pressure. Used by `aero_iv1` for the interstage shoulder; the single-body SV-1 build-up has no compressive turn and so does not exercise it.
aero_nozzle_exit_fraction_GUESS	GUESS. Used only when `area_nozzle_exit` is None and `power_on=True`: the nozzle exit is assumed to occupy $A_e/A_{base} = 0.50$. This is a placeholder for the real value, which WP3's motor model must supply. It changes only the powered-base drag relief.
aero_protuberance_GUESS	GUESS. A single lumped allowance for antennas, umbilicals, fin-root fairings, lugs, joints and surface imperfections, taken as 5 percent of the sum of the other zero-lift components. There is no correlation behind this number. It appears in the breakdown as 'CD_protuberance_GUESS' so it can never be mistaken for a computed quantity.
bulkhead_allowance	GUESS: four ring bulkheads, each a full-diameter disc of 40 percent of wall thickness. Not from a source; a placeholder for joint, rail and lug hardware
case_minimum_gauge	Manufacturing minimum gauge, not a strength requirement. Hoop stress alone gives a sub-millimetre wall at this chamber pressure, which cannot be filament wound or machined. Floors of 1.8 mm (composite, roughly 3 to 4 wound plies) and 1.2 mm (metallic) are taken as engineering minimums. GUESS: no public minimum-gauge standard was found, but without the floor the model returns a motor mass fraction of 0.95, well above the 0.80 to 0.92 band for real tactical motors
insulation_thickness	Sutton and Biblarz, Rocket Propulsion Elements, 9th ed., Chapter 14: EPDM chamber insulation 1 to 5 mm for short-burn tactical motors; 2.5 mm taken here as a mid-range modelling choice
iv1_acs	Attitude-control motor performance. Isp 235 s is the low end of the AP/HTPB range in Sutton and Biblarz, Rocket Propulsion Elements 9th ed., Table 12-1, taken low because a side thruster runs a short, low-expansion nozzle at altitude and pays a large divergence and heat loss. GUESS: the 0.55 inert-to-propellant ratio for a pulsed divert pack is not from a source; it is high relative to a single large motor because a thruster bank pays for multiple valves, nozzles and a manifold. No plume interaction, no minimum impulse bit, no control loop and no roll coupling are modelled.
iv1_aero_fin_station	MODELLING CHOICE, not a measurement: each stage's tail fins are placed with their root TRAILING EDGE at the aft end of that stage, so the stage-1 fins end at the booster base and the stage-2 fins end at the stage-2 to interstage joint. `StageSpec` carries no fin station, and SPEC_IV1.md section 3 shows tail fins at the aft end of each stage. Exposed as the constructor argument `fin_te_gap` so a real gap can be set. The consequence for stability is that the stage-2 fins are about one interstage length further forward in the stacked configuration than the booster fins, which is exactly the geometry the static-margin check needs.

Flagged source	What it says
iv1_aero_strake_Kv	K_v, the vortex-lift factor, is the SUM of a leading-edge and a SIDE-EDGE term, $K_v = K_{v,le} + K_{v,se}$, because a strake is rectangular and unswept: its shed vortex comes off the streamwise tip edge, not off a swept leading edge. This is the Lamar-Polhamus extension of the suction analogy to non-delta planforms: J. E. Lamar, 'Prediction of Vortex Flow Characteristics of Wings at Subsonic and Supersonic Speeds', NASA TR R-428 (1974); J. E. Lamar and B. B. Gloss, NASA TN D-7921 (1975). $K_{v,le} = \pi A/4 = K_p/2$, which TN D-7921 Table III reproduces to better than 0.1 percent for every rectangular entry from $A = 0.05$ to $A = 0.40$. $K_{v,se}$ is taken directly from the TN D-7921 Table III rectangular-wing column (the m1 column, the TR R-428 continuous-loading method that the authors take as the standard), interpolated in A and anchored at $A = 0$ by the analytic limit $K_{v,se} \rightarrow \pi$ stated in TN D-7921 p. 9 and TR R-428 p. 19. See LAMAR_KV_SE_RECTANGULAR for the table. At the default strake pair aspect ratio of 0.043 that gives $K_v = 3.19$. NOTE, because it matters: the value is close to π, NOT to the 2.0 that Bollay's zero-aspect-ratio nonlinear wing theory (ZAMM 19, 1939) gives and that the Newtonian '$2 \sin^2 \alpha$' term in SOURCES['aero_fin_normal_force'] uses. The suction analogy is about 60 percent higher. That is exactly why a strake needs its own method instead of being pushed through the inherited fin routine. APPROXIMATIONS, both of which push the strake normal force DOWN: the interpolation is linear in A, so it is C0 but not C1 with respect to the strake design variables (it is exactly constant with respect to Mach and α, so the trajectory integrator is unaffected); and a body-mounted strake sits in the accelerated cross-flow around the body, where the real vortex is stronger than on the isolated plate the table describes. OMTS vortex breakdown, all Mach dependence, and the strake-to-strake interaction of a cruciform set.
iv1_aero_strake_drag	Strake drag reuses the fin machinery where the physics is identical and says where it is not. Friction: the same Sommer and Short compressible flat-plate law and the same Raymer thickness form factor as the fins, with the Reynolds number on the strake LENGTH, because the strake length is its streamwise chord. Identical physics, so SOURCES['aero_skin_friction'] and SOURCES['aero_fin_form_factor'] apply unchanged. Thickness wave drag: the same Ackeret double-wedge result, SOURCES['aero_fin_wave_drag']. NOT identical physics, and the difference is declared: a real strake is a constant-thickness plate with a short leading-edge bevel, not a double wedge peaking at mid-chord. Concentrating the same thickness into a bevel a few thicknesses long raises the local surface slope by more than an order of magnitude and the wave drag with it, so the double-wedge assumption UNDERSTATES strake wave drag. The opposite end is reported as 'xcheck_CD_strake_le_blunt_bound': stagnation pressure on the whole strake frontal area $n \cdot \text{thickness} \cdot \text{height}$, which is what a fully blunt unbeveled leading edge would cost. The two BRACKET the truth, and at the default geometry the bracket is wide, more than two orders of magnitude, because the double-wedge t/c of 0.0057 spread over a 1.4 m chord is an almost flat surface while a blunt edge is not. Where the real strake falls inside that bracket depends on the leading-edge bevel length, which is NOT an IV-1 design variable, so this model reports the optimistic end plus the bound rather than interpolating with an invented bevel length. Trailing-edge drag: the strake trailing edge is blunt and its base area is charged with the same base-pressure correlation as the body base, SOURCES['aero_base_drag']; that correlation was fitted to an AXISYMMETRIC body base, so applying it to a small two-dimensional trailing edge is an EXTENSION, taken because leaving the term out entirely is the larger error.
iv1_aero_strake_upwash	The Barrowman body-upwash factor $1 + a/s$ (a = body radius, s = strake tip radius) is applied to the strake POTENTIAL term only, reusing SOURCES['aero_fin_body_upwash']. It is a linear potential-flow result, so it is not applied to the vortex-lift term: there is no published upwash factor for a separated-flow load, and the suction analogy already integrates the separated load. APPROXIMATION with a known sign: the real cross-flow acceleration around the body raises the effective incidence of the strake panel, so leaving the vortex term un-augmented makes the strake normal force a LOWER BOUND. Because the potential term is negligible at strake aspect ratios, this choice moves the answer by well under 1 percent.

Flagged source	What it says
iv1_fin_te_at_stage_aft_end	MODELLING CHOICE, and it is visible in the render. SPEC_IV1 gives `StageSpec` a fin semi-span, root chord, taper, sweep and thickness but NO longitudinal station, so the trailing edge is placed flush with each stage's own aft end, which is what 'tail fin' means. On stage 2 that puts the panels immediately forward of the interstage, and because the stage-2 tip-to-tip span (0.56 m on the default) exceeds the booster diameter (0.40 m) the panels stick out past the interstage flare. A real vehicle would move them forward or use a wrap-around skirt. Nothing in the aero or mass model depends on the station, so this is a picture-level simplification, not a numerical one.
iv1_interstage_buckling	The interstage is a compression member: it transmits the full booster thrust into the upper stage. Classical buckling of a thin unstiffened cylindrical shell under axial compression is $\sigma_{cr} = E \cdot t / (R \cdot \sqrt{3 \cdot (1 - \nu^2)})$, which is $0.605 \cdot E \cdot t / R$ at $\nu = 0.33$. Real shells reach only a fraction of that because of imperfection sensitivity. A knockdown factor of 0.30 is applied, which is the conservative end of the 0.15 to 0.50 band quoted for unstiffened monocoque shells in NASA SP-8007, Buckling of Thin-Walled Circular Cylinders. GUESS: 0.30 was chosen inside that band rather than computed from an imperfection survey, and the shell here is conical rather than cylindrical, so the check is an approximation applied at the smaller radius, which is the critical end.
iv1_interstage_mass	MODELLING CHOICE: the interstage is a truncated conical shell between the two stage diameters, of thickness <code>dv.t_interstage</code> in aluminium 7075-T6, plus two ring frames at 40 percent of the shell thickness. Separation hardware (explosive bolts or a linear separation joint, springs or pistons, and the electrical disconnect) is NOT costed. GUESS: that omission makes the jettisoned mass optimistic by an unquantified amount.
iv1_lateral_accel	Available lateral acceleration is the static value $q \cdot S_{ref} \cdot CN_{max} / m$ at the intercept condition, with CN_{max} taken at the alpha limit. APPROXIMATION: it is a capability, not a manoeuvre. It says nothing about autopilot response time, actuator rate or the transient during a turn.
iv1_ogive_polygon_sampling	Inherited, not re-guessed. The revolved outer mould line is a chord polygon, so its volume is exactly the frustum sum that `rocketgen.sizing.masses._tangent_ogive_volume` computes with the same n . WP4 swept n on the SV-1 nose against a 20000-segment reference and measured a total outer-mould-line volume error of -0.106 percent at $n=8$, -0.0117 percent at $n=24$ and -0.0004 percent at $n=128$. The IV-1 stage-2 nose is SHORTER and BLUNTER than SV-1's (1.008 m on a 0.28 m body, fineness 3.6, against 1.05 m on 0.35 m at fineness 3.0), and a chord polygon's error scales with the curvature it has to cut across, so $n = 24$ is at least as accurate here. It is kept unchanged so the two notebooks are comparable.
iv1_qmax	MODELLING CHOICE: 250 kPa dynamic-pressure limit, against 200 kPa for SV-1. A vertical sea-level launch accelerates through dense air, so the same airframe sees a higher peak than SV-1 does launching at 10 km. Consistent with the manoeuvre-envelope pressures quoted for supersonic tactical vehicles in Fleeman, Tactical Missile Design, 2nd ed., Chapter 3.
iv1_qmax_revision	MODELLING CHOICE, and a declared gap. A10 was revised from 250 kPa to 350 kPa because a sweep over propellant, thrust, pitchover angle and pitchover time found a hard floor of about 278 kPa on peak dynamic pressure, rising to 309 kPa for any design that also meets A2, A3 and A4. The structural model does not size the airframe for that load: it is wall thickness times density plus a motor-case hoop-stress check and an interstage buckling check. The airframe mass is therefore optimistic by an amount not quantified here.
iv1_separation	MODELLING CHOICE: stage separation is instantaneous, imparts no impulse and no attitude disturbance, and jettisons the stage-1 inert mass together with the interstage. Real separation carries a tip-off disturbance and a brief drag transient; neither is modelled.

Flagged source	What it says
iv1_strake_mass	Strake mass is the swept plate volume times the titanium density in MATERIALS['fin_t64'], with a 0.85 area-to-volume factor for a tapered leading edge and a root fillet. GUESS: the 0.85 factor is not from a source; a constant-section plate would be 1.0 and a full double wedge 0.5, so it sits between them.
ogive_polygon_sampling	Measured, not guessed. The revolved OML is a chord polygon, so its volume is the frustum sum that `rocketgen.sizing.masses._tangent_ogive_volume` computes with the same n. Sweeping n against a 20000-segment reference on the default SV-1 gives a TOTAL OML volume error of -0.106 percent at n=8, -0.026 percent at n=16, -0.0117 percent at n=24, -0.0066 percent at n=32 and -0.0004 percent at n=128. n=24 is used for the outer profile: 85x inside the 1 percent gate in PLAN.md, and small enough to keep the recipe near 300 blocks.
prop.burn_rate_coefficient	MODELLING CHOICE inside a cited range: reference burn rate 10.0 mm/s at 1000 psia. Sutton Table 12-1 (HTPB/AP/Al) gives 0.25 to 3.0 in/s = 6.35 to 76.2 mm/s at 1000 psia, and the Ch.12 text on p.476 states most composite propellants burn at 7 to 20 mm/s. The coefficient a follows from $a = r_{ref} / p_{ref}^n$.
prop.case_safety_factor	MODELLING CHOICE: safety factor 1.5 applied to material yield strength in the thin-wall hoop-stress case sizing ($\sigma = p * r / t$), plus a 1.0 mm minimum practical gauge. 1.5 is the conventional aerospace pressure-vessel factor; it was not taken from a specific standard in this session.
prop.gamma	ASSUMPTION (not tabulated, treat as a guess inside a physically bounded band): ratio of specific heats of the combustion products taken as 1.20, the value normally used for aluminized AP/HTPB exhaust. Cross-checked: with gamma = 1.20 the cited Isp (262.5 s) and flame temperature (3440 K) imply a mean exhaust molar mass of about 26 kg/kmol, inside the 25 to 30 kg/kmol band of Sutton Figure 12-3 for HTPB-based composite propellant at 68 atm. A +/- 0.02 change in gamma moves the derived c* by about 1.5 %.
prop.ideal_nozzle	APPROXIMATION: frozen-composition, constant-gamma, single-phase isentropic nozzle flow. No two-phase Al2O3 loss, no divergence loss, no combustion efficiency and no throat erosion. Reported Isp is therefore an ideal value; real delivered Isp for a tactical motor of this class is typically 3 to 7 % lower (magnitude of that correction NOT sourced here, so it is not applied).
prop.igniter_mass	GUESS: igniter and safe-arm mass taken as 0.5 % of total propellant mass.
prop.insulation_thickness	GUESS: 3.0 mm of EPDM internal insulation on the case wall and domes. Real insulation is sized from local gas-side heat flux and exposure time, which is out of scope (SPEC.md section 8). Density from MATERIALS['insulation_epdm'].
prop.motor_mass_fraction	CROSS-CHECK CORRELATION, quoted band only, treat the endpoints as a guess: tactical solid rocket motors have propellant mass fractions $m_p / (m_p + m_{inert})$ of roughly 0.80 to 0.91. This band was not verified against a primary source in this session; it is used only to bound the physics-route inert mass, never to replace it silently.
prop.neutral_burning	APPROXIMATION: neutral (constant) burning is assumed within each phase. The burn-rate law is closed at the MEAN web position, which makes propellant mass, web thickness, mass flow and burn time mutually consistent ($w = V_p / A_{b_mean}$, $t_b = w / r$, $\dot{m} = \rho * A_{b_mean} * r$). No burnback simulation is performed, so the reported grain dimensions are volume- and area-equivalent, not a manufacturing drawing.
prop.nozzle_mass_model	GUESS-LEVEL geometric estimate: conical nozzle, 15 deg half angle, 8.0 mm wall, bulk graphite / carbon-carbon density 1800 kg/m^3 (handbook range 1.7 to 1.9 g/cm^3, not verified against a datasheet in this session), multiplied by 1.5 to allow for the convergent section, throat insert and attach ring. This estimate is known to be optimistic; compare against SOURCES['prop.motor_mass_fraction'].

Flagged source	What it says
prop.ramp_times	MODELLING CHOICE, not physics: ignition rise 0.05 s (specified by the WP3 task statement), boost-to-sustain blend 0.10 s, tail-off 0.20 s. The ramps are mass-conserving by construction so total impulse is unchanged.
prop.separation_criterion	Summerfield flow-separation criterion, separation when p_e / p_a is below about 0.4 (Summerfield, Foster and Swan, 1954; the 0.4 figure is the commonly quoted approximation and was NOT verified against the primary paper in this session). Used only to raise a warning, never to change a force.
prop.terminal_grain	MODELLING CHOICE: the terminal segment is an internal-burning tubular grain, the same geometry and the same mean-web closure as the boost segment, because an end burner at this bore cannot supply the burning area a multi-kilonewton terminal thrust needs at a credible chamber pressure. Its length counts towards the SPEC.md section 4 grain L/D limit of 1.0 to 8.0.
prop.terminal_pulse	MODELLING CHOICE with one guess: the terminal pulse is a third grain segment in the same case, ignited on command rather than on a timer, which is how real dual-pulse tactical motors work (an inter-pulse bulkhead separates the charges and a second igniter fires the aft one). Its default chamber pressure is $1.0 \times \text{DesignVector.p}_c$, that is the pressure the case is already sized for, so no new number is introduced. GUESS: the inter-pulse bulkhead, its insulation and the second igniter are NOT costed as separate inert mass; the terminal grain is charged only the same 0.5 % igniter fraction as the rest of the motor, so the reported inert mass is optimistic for a pulsed motor. Physical motivation: an unpowered terminal dive is terminal-velocity limited and cannot reach Mach 1.5 at sea level for any dive angle, so thrust in the endgame is the only way to meet SPEC R6.
prop.two_position_throat	MODELLING CHOICE: the motor is given a two-position throat (ejectable boost throat insert) so the boost and sustain phases can run at different throat areas with the same nozzle area ratio. Without it, a single throat sized for the boost thrust drives the sustain chamber pressure to about 0.13 MPa and Kn to about 22, which no composite propellant will sustain. Set <code>two_position_throat=False</code> to model one fixed throat instead.
prop_iv1.booster_aft_closure	MODELLING CHOICE: the booster grain bay is the stage length less a 0.10 m aft closure, nozzle attach ring and igniter boss allowance. The value is not new: it is the same <code>CASE_LENGTH_ALLOWANCE_M</code> that <code>propulsion.py</code> already adds to the grain length to get the case length, so bay length and case length are consistent with each other. The payload stage instead gets $L - L_{\text{nose}} - L_{\text{seeker}} - L_{\text{payload_bay}}$, which is set by the internal layout in <code>config_iv1.StackDesignVector</code> .
prop_iv1.fin_mass_model	GUESS, carried over unchanged from the analytic branch of <code>masses.py::build_masses</code> so the two modules cannot disagree about the same hardware: fin mass is $n_{\text{fin}} \cdot S_{\text{fin_exposed}} \cdot t_{\text{fin}} \cdot 0.65 \cdot \text{density}(\text{Ti-6Al-4V})$, where 0.65 is an area-to-volume factor for a tapered biconvex section. No source was found for the 0.65; it is a shape factor chosen to be below the 1.0 of a flat plate.
prop_iv1.interstage_shell	MODELLING CHOICE: the interstage is charged as a cylindrical shell at the BOOSTER diameter, of length <code>StackDesignVector.L_interstage</code> and thickness <code>t_interstage</code> , in <code>MATERIALS['airframe_al7075']</code> . Mass is then exactly $\pi \cdot D1 \cdot L \cdot t \cdot \rho$, linear in <code>L_interstage</code> . The conical frustum from <code>D2</code> to <code>D1</code> is computed alongside it as a cross-check and is lighter, because a taper has less lateral area than a cylinder of the same axial length; the heavier cylindrical value is the one charged. GUESS: no buckling or crush-load sizing is performed, so <code>t_interstage</code> is an input, not a result.
prop_iv1.p_c_min_reliable	GUESS: 1.0 MPa taken as the chamber pressure below which composite-propellant combustion becomes unreliable, so a stage below it is warned about. The figure is the same one <code>propulsion.SolidMotor</code> already warns on and no primary source for it was verified in this session. It changes no force and no mass: it only raises a warning. Every IV-1 stage runs at <code>StageSpec.p_c</code> , which defaults to 8.0 MPa, so the check is inactive for the default stack and exists for design vectors the sizer may reach.

Flagged source	What it says
prop_iv1.separation_hardware	GUESS, and an admitted omission: the separation joint itself is NOT costed. A real tandem separation needs a linear-shaped-charge or clamp-band joint, springs or retro-rockets, and ring frames at both ends of the interstage. None of that appears in jettisoned_mass(), so the reported jettisoned mass is the shell and motor hardware only and is optimistic by an amount this model cannot quantify.
prop_iv1.separation_timing	MODELLING CHOICE, following the deliverable contract: stage 1 ignites at $t = 0$ and burns to the end of its tail-off, the stack then coasts unpowered for <code>InterceptRequirements.t_coast_separation</code> , and separation and stage-2 ignition are the same instant, $t_{\text{separation}} = t_{\text{burnout}}(1) + t_{\text{coast_separation}}$. SPEC_IV1.md section 5 allows a short unpowered coast between boost and stage-2 ignition and does not fix whether the jettison happens at the start or the end of it; putting it at the end means the stack carries the booster mass through the coast, which is the pessimistic and therefore the safe choice for the trajectory.
prop_iv1.stage1_sized_at_sea_level	MODELLING CHOICE, following the StageSpec.F_thrust definition in config_iv1.py: stage-1 thrust is a SEA-LEVEL equivalent, because SPEC_IV1.md requirement A1 launches the stack vertically from a canister at sea level. The stage-1 throat is therefore sized with $p_a = 101325$ Pa. Upper-stage thrust is a VACUUM value and the upper-stage throats are sized with $p_a = 0$, because they are lit above 10 km. Both use the same relation $F = C_F \text{vac} * p_c * A_t - p_a * A_e$; only the sizing p_a differs.
prop_iv1.stage_is_a_phase	MODELLING CHOICE: one stage of the IV-1 stack is modelled with exactly the same closure as one phase of the SV-1 dual-thrust motor, that is one propellant charge at one chamber pressure through one fixed throat, sized by <code>propulsion.design_phase()</code> . The physical difference between a phase and a stage is not in the internal ballistics but in the hardware: a stage brings its own case and nozzle, and stage 1 leaves the vehicle at separation.
prop_iv1.tubular_stage_grain	MODELLING CHOICE and a known conservatism: every stage grain is an internal-burning tube closed at its mean web, the same geometry as the SV-1 boost segment. A tube is the least area-efficient internal-burning geometry. A slotted, star or finocyl grain reaches several times the burning area in the same case length, which is how real boosters of this thrust class stay short. No shape factor is applied here, because no sourced multiplier was available; the model reports the long tube and its L/D instead. A stage that fails the SPEC_IV1.md section 6 L/D limit of 1.0 to 8.0 with a tubular grain may still close with a shaped grain, and the warning says so.
radome_thickness	GUESS: 6 mm Pyroceram radome wall. No public thickness standard was found for a 0.35 m class radome; scaled from the requirement that wall thickness be an integer number of half-wavelengths at Ku band, which is not modelled here
traj.alpha_limit	MODELLING CHOICE: trim angle of attack is limited to 15 deg. Trim demands beyond that are recorded as an authority shortfall rather than flown, so the sizing loop can see them. 15 deg is a conventional tactical-rocket limit; it was not taken from a specific source in this session, so treat it as a guess.
traj.dive_entry	MODELLING CHOICE: SPEC.md section 2 does not define the coast-to-dive transition. Rule 'max_range' (default): hold altitude until level trim can no longer be held inside the alpha limit, or until Mach falls to 1.20, then dive at <code>Requirements.gamma_terminal</code> . Rule 'range': dive as soon as the current range plus the geometric dive range reaches <code>Requirements.range_min</code> . Rule 'terminal_boost': dive the moment the sustain phase ends, with no level coast at all, which keeps the most kinetic and potential energy for the endgame.
traj.gravity	Constant gravity $g = 9.80665 \text{ m/s}^2$ (standard gravity, CGPM 1901). MODELLING CHOICE: the inverse-square variation is neglected, which is under 0.4 % over the 0 to 12 km band of this mission and keeps the analytic validation cases in tests/test_trajectory.py exact.

Flagged source	What it says
traj.guidance_gains	MODELLING CHOICE, no source: the altitude-hold and flight-path-angle autopilot gains (0.15 1/s outer, 2.0 1/s inner), the 200 m/s commanded climb rate cap, the 45 deg commanded climb angle cap and the 20 g commanded load factor cap are arbitrary. Guidance law design is an explicit non-goal (SPEC.md section 8); these gains only have to fly the SPEC.md section 2 profile smoothly. They do not appear in any reported performance number other than through the flown trajectory. The load factor cap matters for one thing only: it keeps the recorded CN_required history finite at the dive-entry command step, so the sizing loop can read it.
traj.terminal_boost_ignition	MODELLING CHOICE: SPEC.md section 2 has no terminal-boost phase, so the ignition trigger is invented here and exposed rather than hidden. Rule 'carry_to_impact' (default) lights the pulse when the altitude falls to $\text{terminal_ignition_margin} * V * \sin(\gamma_{\text{terminal}}) * t_{\text{terminal}}$, that is when the predicted remaining descent time equals the terminal burn time, so the pulse burns out at impact instead of above the ground. Rule 'dive_entry' lights it at dive entry. Rule 'never' leaves it unlit, which is what the two-phase regression case uses. The margin defaults to 1.0 and is a free parameter for the sizing loop: the prediction uses the CURRENT speed and the pulse accelerates the rocket, so a margin near 1.0 lights slightly late and the pulse is still burning at impact. That is the safe side; a margin above about 1.3 wastes propellant above the ground.
traj_iv1.launch_speed	GUESS: the vertical rise starts at 1.0 m/s rather than at rest. A 3-DOF point mass at exactly zero speed has an undefined flight-path angle and a degenerate γ_{dot} equation, so the speed must be non-zero. No canister exit-velocity figure was sourced. The choice is close to harmless: at the IV-1 launch thrust-to-weight the vehicle passes 1 m/s within about 10 ms of ignition, so the programme is shifted by that much and the burnout speed changes by under 1 m/s. Exposed as the AscentMission($v_{\text{start}}=...$) argument so it can be traded.
traj_iv1.mass_programme	The mass programme is owned by the integrator, not by the mass statement: $m(t) = m_0 - \text{propellant burned} - \text{jettisoned mass}$. 'mass_stack' supplies only m_0 , read from a float, from 'total_mass' (config.MassStatement), from 'total' (masses.MassBuildup), from 'm0', from 'launch_mass', from a zero-argument callable, or from mass_at(0, 0, False). Nothing is guessed and no default launch mass exists: an unrecognised object raises.
traj_iv1.pitch_programme	MODELLING CHOICE, and SPEC_IV1.md section 8 requires it to be declared. The vehicle rises vertically until $dv.t_{\text{pitch}}$, then turns at the constant commanded rate $dv.\text{pitch_rate_max}$ until the flown flight-path angle reaches $dv.\gamma_{\text{pitch}}$, then flies a pure gravity turn (zero commanded normal force) for the rest of the flight. There is NO autopilot gain: the commanded rate is a design variable applied open loop, and the end of the turn is found by bisecting the step that crosses γ_{pitch} . There is no thrust-vector control and no side thruster, so the turn is flown on aerodynamic normal force alone and is frequently limited by the alpha limit; the limited steps are flagged rather than flown as commanded.
traj_iv1.q_floor	MODELLING CHOICE, carried over from rocketgen.sizing.trajectory: below 200 Pa of dynamic pressure the required-normal-force coefficient is frozen at zero instead of being divided by a vanishing $q * S_{\text{ref}}$. Reaching this on a lofted arc is recorded in TrajectoryResult.message and in diagnostics['q_floor_hit'].
traj_iv1.velocity_floor	GUESS: the run is declared stalled below 20 m/s. Nothing in SPEC_IV1.md sets a floor. It exists because the γ_{dot} equation divides by speed, so an arbitrarily slow point mass is numerically meaningless, and because the aero model is not valid there either. The check is ARMED only after the vehicle has once exceeded the floor, so the 1 m/s launch condition cannot trip it. Exposed as AscentMission(velocity_floor=...).

Flagged source	What it says
wave_drag_applied_as_ratio	MODELLING CHOICE, stated rather than hidden. Slender-body theory is used only for the RATIO between two shapes at the same length and base area, multiplying the Bonney correlation which keeps the Mach dependence and the level validated against the 23 Basic Finner shots. The absolute slender-body drag level is NOT summed into CD0. At the tangent-ogive shape the ratio is 1.0 to machine precision, so the pre-spline results are reproduced exactly.
wave_drag_mach_independence	Leading-order slender-body wave drag carries no Mach number. That is a property of the theory, not an omission. Because it is applied here as a ratio against an ogive evaluated at the same Mach, the Mach dependence of the delivered CD_wave_body is entirely Bonney's. GUESS-FREE but APPROXIMATE: the true shape sensitivity does drift with Mach number, and that drift is not modelled and has not been quantified here.

Table 14. All 57 flagged sources. Two belong to the work this report describes: the entry that records the wave-drag ratio as Mach-independent, and the entry that records it as applied as a ratio.

What this study does not claim

- The drag calibration is not applied to IV-1. Section 7 states this in full. It is an open defect, not a modelling choice.
- The slender-body wave-drag ratio is Mach-independent at this order. The true shape sensitivity drifts with Mach number. That drift is not modelled and has not been quantified.
- The nose shape was SET to the drag optimum, not searched. This study therefore makes no claim about an interior optimum on this vehicle. The SV-1 spline study, where the shape was searched, found one.
- The nozzle model is ideal. There is no two-phase loss, no divergence loss, no combustion efficiency and no throat erosion. Real delivered specific impulse for this class runs 3 to 7 percent lower and that penalty is not applied.
- The grains are restricted to a tubular geometry. Real motors of this class use finocyl or star grains. The restriction caps booster thrust and holds stage-2 propellant down. It is a visible design constraint here rather than a hidden optimism.
- Stage-2 static margin is still not in the constraint list. That remains open from the tangent-ogive IV-1 study, where it was evaluated once for the report and failed.
- Six-degree-of-freedom flight mechanics, guidance law design, structural sizing beyond a wall-thickness and hoop-stress check, and CFD are all out of scope.
- The requirements are invented for the demonstration. They do not correspond to any real programme.

Warnings the run raised, and did not swallow

Warning
stage 1 motor inert mass raised from a bottom-up 13.9 kg to the tactical-motor correlation floor 21.8 kg; the bottom-up model is incomplete by 7.9 kg. See SOURCES['prop_iv1.stage_inert_incomplete'].
stage 2 motor inert mass raised from a bottom-up 7.6 kg to the tactical-motor correlation floor 8.9 kg; the bottom-up model is incomplete by 1.3 kg. See SOURCES['prop_iv1.stage_inert_incomplete'].
stage 1: motor inert raised from 13.9 kg to the correlation floor 21.8 kg
stage 2: motor inert raised from 7.6 kg to the correlation floor 8.9 kg
interstage 3.02 kg (shell 2.37, rings 0.64); separation hardware is not costed

Table 15. The 5 warnings the converged spline run raised. The tangent-ogive run raised the same set, so none of them is caused by the shape change. The bottom-up motor inert model is incomplete by design, and the correlation floor books the shortfall as a visible line item rather than hiding it.

9. Reproducing this result

```
.venv/Scripts/python.exe -m pytest tests -q
.venv/Scripts/python.exe scripts/iv1_converge.py --nose spline --interstage spline
.venv/Scripts/python.exe scripts/iv1_converge.py --nose tangent_ogive --interstage cone
.venv/Scripts/python.exe -m rocketgen.report.evidence_iv1_spline
.venv/Scripts/python.exe -m rocketgen.report.fig_iv1_spline
.venv/Scripts/python.exe -m rocketgen.report.build_report_iv1_spline
```

The evidence collector does NOT call nTop. Both stacks were already measured, and the per-body measurements are on disk. Those are read back and fed to the mass and aerodynamic models exactly as the sizing script fed them. Every re-flight is then checked against the recorded result, field by field, to a relative tolerance of 1.0e-9. A re-flight that did not reproduce its record fails the collector rather than quietly publishing a second answer.

Component	Version or value
Python	3.11.15
numpy	1.26.4
scipy	1.13.1
SUAVE	2.5.2, vendored from github.com/suavecode/SUAVE
nTop Automate	5.53.2 installed, 5.54.0 development build
Drag calibration available	1.171
Drag calibration applied	NO
Evidence collection wall time [s]	29.9

Table 16. Environment. numpy and scipy are pinned below their current major versions because SUAVE 2.5.2 does not run on either.

Two toolchain warnings are worth repeating. The nTop command-line tool returns exit code 72 when a block fails, not on success. Widely repeated guidance says the opposite and is wrong. Judge success on the expected files existing and being non-empty. Second, the cost of a surface-area measurement tracks the complexity of the implicit field, not the size of the body. On this vehicle four measurements on booleaned bodies took about 24 s each, and the fifth, on a bare cylinder primitive, took 0.27 s for the largest area of the five.

A note on the renders in Figures 1 and 2. The first version of this report had no picture of the vehicle at all, because neither IV-1 run turned on the mesh export and no STL existed to render from. That is not recoverable after the fact: it needs the measurement repeating with exports enabled, which scripts/render_iv1_spline.py now does at the converged point only. A report without a view of its own geometry cannot be checked by a reader, so this is now a standing rule of the repository rather than a per-report decision.